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Radiation Protection in Medicine

Radiation protection aims to reduce unnecessary radiation exposure with a goal to minimize the harmful effects of ionizing radiation. In the medical field, ionizing radiation has become an inescapable tool used for the diagnosis and treatment of a variety of medical conditions

The background radiation is contributed mainly by several sources, Terrestrial radiation, cosmic radiation, and radiation from radioactive elements in our bodies. Terrestrial radiation varies based on surrounding materials, including buildings (granite rocks contain small amount of Uranium-238 producing radon).
Cosmic radiation levels change with elevation and latitude , the background at 3000m attitude is about 20% higher than that at sea level. The internal irradiation arises mainly from potassium 40 (40K)in our body, which emits gamma ray and beta- rays and decays with a half life of 1.3x109 years.
The total effective dose equivalent for a member of the population in the United States from various sources of natural background radiation is ~ 3.0 mSv/ year (300 mrem/year).

Radon is a naturally occurring radioactive gas, recognized as being carcinogenic, and is present everywhere on the surface of the Earth. As we breathe, some of these radioactive dust particles stick to the lining of the lungs, thus the lung receive more radiation than the rest of body. The dose equivalent to the lungs is about nine times the amount to the rest of the body. Radon is one the cause of lung cancer behind smoking. Joint exposure to radon and tobacco substantially increases the risk of developing lung cancer.

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Biological effects of ionizing radiation

Biological effects of ionizing radiation are of two general types somatic and genetic.
Somatic effects affect an individual directly (loss of hair , reddening of the skin , etc.). Somatic effects depend on the amount of radiation received, the part of body irradiated ,and the age of patient. Younger person, the more hazardous the radiation . The most dangerous period to receive radiation is before birth. At certain period in the development of the fetus, radiation can produce deformities.

Figur (1): Litter from a female mouse irradiated with x-rays during organogenesis and sacrificed at 19 days. Several anomalies are demonstrated in this litter. There are four resorbed embryos **(bottom)** and five fetuses that would have been born alive **(top)**. From left to right, the two on the left both have brain hernia, and second from the left has the gut outside the body. The two on the right have only partially developed heads, only one at middle is apparently normal. The four objects below were killed as embryos by x-rays.

FIGURE(1): An entire litter of nine mice irradiated with x-rays during organogenesis and sacrificed at 19 days. (Photograph by Dr. Roberts Rugh.)

Somatic effects of radiation are the reddening of the skin , loss of hair , stiffening (fibrosis) of the lung , reduction of white blood cell (leukemia) and the induction of cataracts eye.

The severity of many somatic effects is carcinogenesis, the induction of cancer. It has been found that radiation can induce many types of cancer besides skin cancer. Radiation to the thyroid has caused thyroid cancer, and radiation to the blood –forming organ (bone marrow) has caused blood or leukemia.
Genetic effects consist of mutations in the reproductive cells that affect later generations. Since genetic effects occur only when reproductive cells are irradiated, the gonads should be shielded during x-ray studies when possible.

In order to evaluate the genetic effects of x-rays on population, the concept of the genetically-significant dose (GSD) is useful. The GSD due to an exposure depends on the dose to the ovaries and testes as well as the age of the individual, which determines the probability of that person becoming a parent in subsequent years. Thus x-raying women over 50 years old, who normally have little chance of having babies, contributes very little to the GSD of the population. Exposure of the reproductive organs of children results in the maximum contribution to the GSD of the population, since their potential for producing offspring is at maximum.
In the United States, x-raying young men results in a major contribution to the GSD. An x-ray that includes the testes, which have little shielding tissue, results in a larger genetic dose than a similar x-ray that includes the ovaries. On the other hand, it is easy to shield the testes without losing any diagnostic information, while this is not true for the ovaries.

Radiation Protection Units and limits

The Becquerel(Bq) is the SI unit for radioactivity (activity), counts how many particles or photons (in the case of wave radiation) are emitted per second by a source. The absorbed dose gray (Gy) or rad refers to the amount of energy deposited by ionizing radiation in a given material, typically biological tissue.

Dose Equivalent (DE): in radiation refers to a measure of the biological effect of ionizing radiation on living tissue

$$\text{Dose Equivalent (DE)} = \text{rad QF}$$

The QF takes into account the increased damage done by certain types of radiation. A rad of densely ionizing radiation dose much more damage to a cell than a rad of the x-ray , gamma, or beta rays and is assigned a larger QF.

QF is related to the relative biological effectiveness (EBE) . Both QF and (EBE) are due to the increased biological effects of densely ionizing radiation. Table (1) lists the QF for varies types of radiation . The QF for x-rays , gamma rays, and beta rays is unity, so for these types of radiation rads equal remes.

The amount of man-made radiation to the public is about 0.17 rem / year. Individual member of the public may receive 0.5 rem/year , and radiation worker are allowed 5 rems/ years..

Table 1: The quality factors for various type of radiation

Type of radiation	QF
x- and g-rays, electrons and positrons energies > 0.03 MeV	**1**
electrons and positrons energies < 0.03 MeV	**1**
Neutrons energy < 10 keV	**3**
Protons	10
Alphas particles	**20**

Radiation protection instrumentation:

Radiation monitoring instruments are used both for area monitoring and for individual monitoring. The instruments used for measuring radiation levels are referred to as area survey meters (or area monitors) and the instruments used for recording the dose equivalents received by individuals working with radiation are referred to as personal dosimeters (or individual dosimeters). All instruments must be calibrated in terms of appropriate quantities used in radiation protection.

A- Area Monitoring instruments:

1- Ionization chambers counters: This is a sealed cylindrical air-filled chamber, the weak current is amplified and displayed on a meter . The Ionization chambers has a good response for beta- ray , x-ray and gamma-ray , it is used for high radiation levels ranging from 0.01 to 5 R/hr. Two version of device are available , one must be placed in a (reader) , and other has a small electrometer and magnifier can be read directly while holding it to the light, it does not accurately measure x-ray in the diagnostic region. The maximum reading of a pocket ionization chamber is generally 0.2 R . Fig (1)

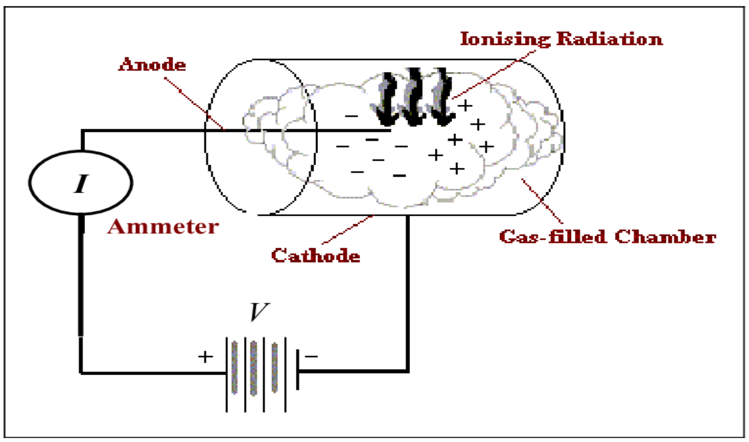

Figure 1: the portable ionization chamber

2- Geiger-Müller counter (G-M tube)

It is much more sensitive than ionization chamber due to gas multiplication • GM detector gives a count for each ionizing event it detect ,whether event is low or high energy , typically count 200 to 300 counts per minute. GM provide an audible chirp for every few hundred counts, useful as personnel monitors they quickly alert when there is increase in the radiation field Fig (2).

FIGURE (2): A pocket Geiger- Mueller

B- Individual monitoring:

Individual monitoring is used for those who regularly work in controlled areas or those who work full time in supervised areas. The most widely used individual monitoring systems are based on either TLD dosimetry or film dosimetry.

1-   A film badge

The film are used for workers radiation safety, the badge holder creates a distinctive pattern on the film indicating the type and energy of the radiation received. The badge incorporates a series of filters to determine the quality of the radiation. Radiation incident on the badge will produce a different degree of darkening under each filter. They can measure beta- ray dose as well as x-ray and gamma-ray. Disadvantages of a film badge responds better to low energy than x-ray than to high energy gamma ray , and in warm humid environment , the signal fades rapidly (50% in 1 week), as shown in FIG(3).

Figure (3): The Film badge

2- **Thermoluminescent dosimeter(TLD):**

The way that a TLD dosimeter measures this exposure is through a crystalof Lithium fluoride (LiF) that is embedded within its detector. This crystal emits a visible light when it is heated by radiation, and the more radiation exposure that the TLD crystal receives, the more intense the light that the TLD crystal emits will be. This type of TLD dosimeter is mostly seen in use for monitoring the beta and gamma radiation that is emitted from x-ray machines.

A Thermoluminescences are used for measuring radiation to patients undergoing radiation therapy treatment , the dosimeters are placed inside body cavities such as the bladder and rectum (FIG: 4) .

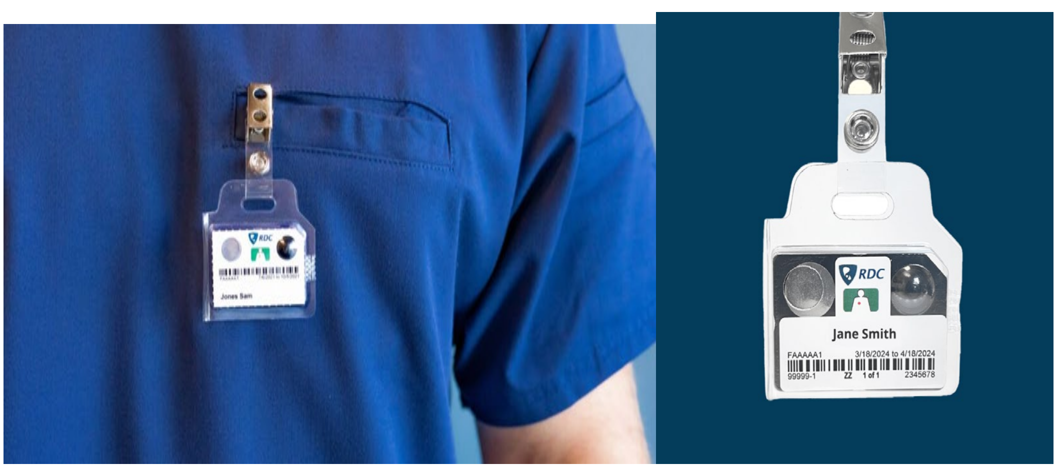

Figure (4): Thermoluminescent dosimeter(TLD)

3- Quartz fiber dosimeter

Is a type of radiation dosimeter, a pen-like device that measures the cumulative dose of ionizing radiation received by the device, usually over one work period. It is clipped to a person's clothing, normally a breast pocket for whole body exposure, to measure the user's exposure to radiation. The device is mainly sensitive to gamma and x-rays, but it also detects beta radiation above 1 MeV. Neutron sensitive versions have been made. exposure ranges usually

Measure up to (5 mSv).

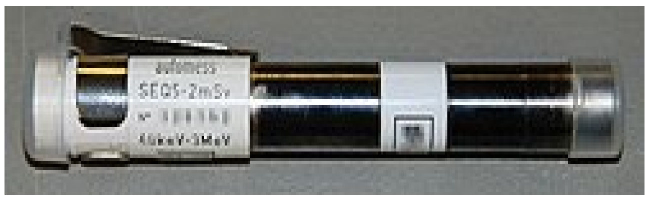

Figure (5): Quartz fiber dosimeter**:**

4- The electronic personal dosimeter (EPD)

Is a modern electronic dosimeter for estimating uptake of ionizing dose of the individual wearing it for radiation protection purposes. Has continuous monitoring which allows alarm warnings at preset levels and live readout of dose accumulated. It can be reset to zero after use.

Figure (6): The electronic personal dosimeter (EPD)

<u>Radiation protection in diagnostic:</u>

Ionising radiation is used in medicine for main diagnosis applications which is mainly imaging – such as: X-ray imaging like Computed Tomography (CT) which gives anatomical images and gamma camera to image body functions

Medical x- rays are the largest source of man –made radiation. Several factors contribute to larger exposures during taking x-rays for diagnostic.

If the x-rays unit does not have sufficient filtration in the beam , the low energy x-ray that would be removed by filter enter the body and contribute to the patient exposure . Most x-ray units should have at least 3 mm of aluminum filtration.

Because of the importance of reducing genetically significant radiation , the gonads be shielded whenever possible. In taking x-ray of infant chests, it is desirable to shield the entire lower abdomen.

In fluoroscopy as in conventional radiology , there should be sufficient filtration in the beam of x-ray to remove the low-energy x-ray and radiation beam should be collimated to the area of interest . An x-ray beam larger than necessary produce more scatter than a beam of the proper size, and since scatter seriously degrades the image, the patient get more radiation .

New electronic image devices should help reduce radiation from fluoroscopic examination, the amount of radiation needed to produce one electronic image is about 1% of that needed to produce an image with the conventional screen- film combination .The quality of image is not as good as with film , but it is satisfactory for many situation and the image is available immediately .

A rapmeter is a flat ionization chamber as shown in (Figure 7), that is attached to the end of the collimator, it does not touch the patient or interfere with the operation of x-ray unit. If all x-ray units and fluoroscopic units had rapmeters attached , x-ray operators become more aware of the amount of radiation they are given to patients.

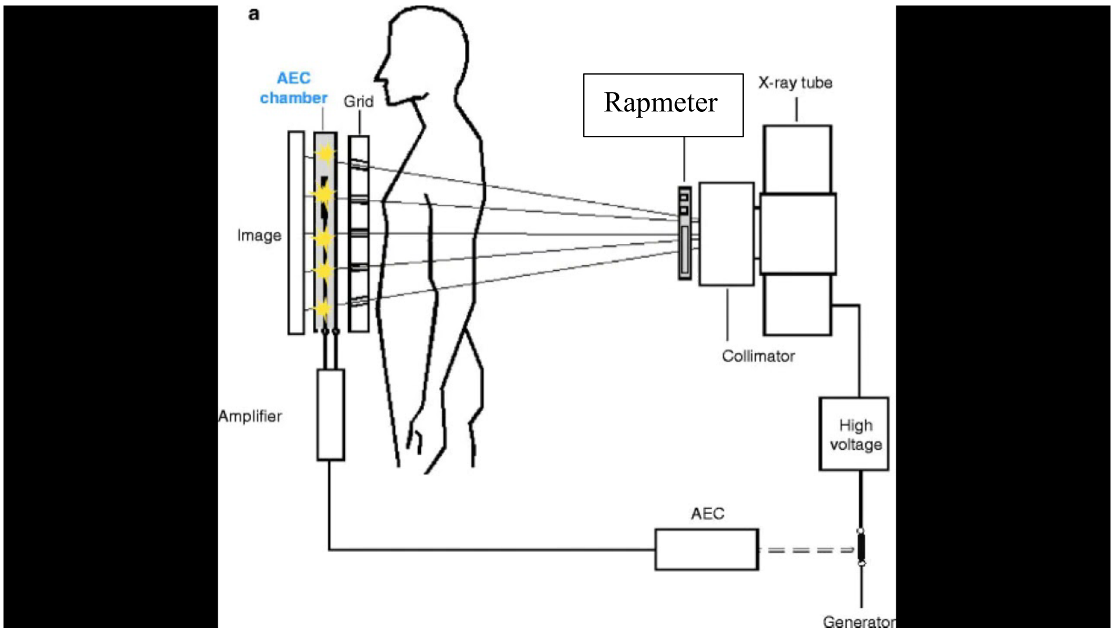

Figure (7): the principle of rapmeter

Radiation protection in radiation therapy

Therapy is treatment of diseases (mainly cancer) using, for example:

- External beams of gamma rays, protons etc
- Small sealed sources placed next to the cancer

Since the radiation therapy area of a hospital contains intense radiation sources, it is typically surrounded by concrete walls. Treatment room and surrounding areas about 0.5 m thick, Also , the scattered radiation during a radiotherapy treatment is large , to protect individuals who might enter the room during treatment the door has a switch (interlock) that turns off the machine when the door is open .

The radiation sources themselves must be adequately shielded.   The gamma radiation from Cobalt 60 source shielded to reduced gamma ray to the safe level .